



DEPARTMENT OF INFORMATION TECHNOLOGY
CENTER FOR DIPLOMA STUDIES
DAT 21101 DIPLOMA PROJECT I (PROPOSAL)
Session/Semester 2025/2026 Semester 1

SECTION A: STUDENT DETAILS

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SECTION B: PROJECT INFORMATION

Supervisor's Name : CIK FAUZIAH BINTI ABDUL RAZAK @ LAPOCHIK

Project Title : Predictive Animal Warning System

Background:

Human–wildlife and stray animal invasions into residential, villages, and urban areas have escalated significantly, causing damage to property, killing of livestock, and threatening human safety. Most of the time, security devices like motion detectors and CCTV cameras are utilised to detect human intrusions, meaning that they cannot distinguish animals give alerts to false alarms. There is evidence that an IoT-based animal detection and monitoring system can be a reliable source of information on wildlife activities. In the recent past, a work had proposed an IoT sensor network for wild-animal detection along roads to prevent vehicle collisions, showing the possibility of combining microcontrollers, sensors, and wireless communication. It demands a permanent network infrastructure. Another research introduced an IoT-based wildlife monitoring system that sends GSM notifications when animals are detected. Although, it does not have the capability to make predictions or adapt locally for non-highway areas. With the help of image recognition, sensor data, and GSM-based communication, the proposed Predictive Animal Warning System (PAWS) is a step towards solving these problems. With the help of image recognition and sensor data, the proposed system is designed to provide timely alerts to mitigate the risk of animal-vehicle collisions on high-risk roadways, such as Jalan Pagoh - Batu Pahat. The system, powered by an IoT platform like Blynk, will send real-time data, captured images, and logs to supervising authorities like the Police, while simultaneously activating a blinking light alert system for immediate driver warning.

References:

- [1] “IoT Sensor Network for Wild-Animal Detection near Roads,” *Sensors*, vol. 23, no. 21, p. 8929, 2023. <https://www.mdpi.com/1424-8220/23/21/8929?utm>
- [2] A. S. Sree, K. Deepthi, N. Kathika, and S. Santhosh, “IoT-Based Wildlife Monitoring System,” *Int. J. Res. Publ. Rev.*, vol. 4, no. 5, pp. 2111–2114, May 2023. <https://www.ijraset.com/best-journal/beast-sentinel-smart-iotbased-wildlife-tracking-and-intrusion-system?utm>
- [3] D. Ranparia, A. Patel, and H. Mehta, “Wild Animal Intrusion Detection System Using YOLO,” *Int. J. Innov. Sci. Res. Technol.*, May 2023. <https://ijisrt.com/assets/upload/files/IJISRT23MAY1948.pdf?utm>
- [4] M. Nizam Osman, M. H. Faeq Ismail, K. A. Sedek, N. Arzami Othman, M. Maghribi, “A low-cost home security notification system using IoT and Telegram Bot: design and implementation,” *Journal of Computing Research and Innovation (JCRINN)*, vol. 7, no. 2, pp. 327-337, 2022. <https://ir.uitm.edu.my/id/eprint/69025/?utm>

Objectives:

1. To design and develop an Admin and user mobile interface using Blynk for Predictive Animal Warning System (PAWS), enabling monitoring of real-time background data, viewing detection history logs, and access to captured images by authorities.
2. To evaluate the effectiveness of PAWS in providing timely alerts and notifications through Blynk and implement a supplementary blinking light system to provide immediate, visual warnings to drivers.
3. To implement and test the PAWS prototype in collaboration with the Police along the high-risk segment of Jalan Pagoh - Batu Pahat, to evaluate its reliability in detecting wild animal presence and providing timely alerts.

Scope & Significance:

Scope

1. The proposed system will be deployed and tested in collaboration with the Police along a designated high-risk segment of Jalan Pagoh - Batu Pahat to evaluate its reliability in detecting wild animal presence and providing real-time alerts.
2. The system will utilise a Raspberry Pi as the main processing unit, integrated with camera sensors for object detection, a blinking light component for immediate driver alerts, and the Blynk platform for sending instant, monitored data and notifications to authorities
3. The proposed system is designed to monitor an approximate detection area based on the camera's field of view (FOV), typically ranging between 60° to 120°, depending on the placement and type of camera module used. The effective detection range may vary depending on environmental conditions and camera positioning within its protective enclosure.

Significance

1. **Improved Safety**
Detects and alerts users of nearby wild animals, reducing risks of harm or property damage in rural and residential areas.
2. **Reliable Communication**
Utilises the Blynk IoT platform to ensure police/authorities receive timely alerts through logged alerts and an accessible messaging system.
3. **Support for Wildlife Awareness**
Helps users and communities understand animal movement patterns, encouraging safer coexistence between humans and wildlife.

Methodology:

The development of the Predictive Animal Warning System (PAWS) will be carried out through several structured phases to ensure the system functions efficiently and meets the stated objectives. The workflow of the project is divided as follows:

Phase 1: System Development

This phase focuses on developing a functional prototype capable of detecting wild animals and issuing alerts through Blynk.

- Develop a two-tier Admin/User interface on the Blynk platform for data logging, history review, and camera feed monitoring with Blynk integration.
- Integrate and program the blinking light module, ensuring its tempo is designed to alert drivers without causing undue disturbance.
- Choose suitable hardware such as Raspberry Pi, camera module, and required sensors.
- Develop the detection program using the Python language, incorporating OpenCV or TensorFlow Lite for animal recognition.
- Configure the Blynk API to enable real-time Data logging and message alerts to authorities.

Phase 2: System Implementation and Testing

The second phase emphasizes real-world implementation and initial testing.

- Ensure testing and evaluation focus on the road environment, the effectiveness of the blinking light, and the reliability of the Blynk data logging.
- Test camera positioning, detection range, and field of view while keeping the device safely enclosed.
- Observe system performance under various lighting and movement conditions.
- Verify that Telegram notifications are delivered promptly when animal movement is detected.

Phase 3: System Evaluation and Optimization

This phase involves evaluating PAWS in terms of accuracy, reliability, and user responsiveness.

- Record detection events and Blynk alert logs to measure system accuracy and false detection rates.
- Assess overall performance in identifying animals within its effective field of view.
- Evaluate system stability, processing speed, and alert delay times.
- Make adjustments to improve detection accuracy, image processing speed, and notification reliability.

Phase 4: Documentation and Review

The final phase focuses on compiling the full documentation of the PAWS development and testing processes.

- Prepare a detailed report covering system design, implementation, test results, and evaluations.
- Discuss limitations and provide recommendations for future improvements such as expanding coverage, adding more sensors, or improving camera protection.
- Present the final findings and project outcomes.

Expected Outcome:

Expected Results (PAWS: Predictive Animal Warning System) aims to:

1. **Development of an Automated Animal Detection System**
 - A fully functional PAWS prototype capable of detecting wild animals using camera and sensor data, and automatically sending alerts to users through Blynk platform interface (Admin/User) and activating a physical blinking light alert system.
2. **Successful Field Implementation**
 - The PAWS system is successfully tested in along Jalan Pagoh - Batu Pahat in collaboration with the Police, proving its reliability and adaptability in real environmental conditions.
3. **Evaluation of System Accuracy and Efficiency**
 - The system demonstrates accurate detection performance, timely Blynk notifications, and stable operation, enhancing safety and awareness for users living in areas exposed to wildlife risks.

****Selected Application Development Tools:**

To develop this system, a combination of hardware and software tools will be used to ensure accurate animal detection, reliable alerts, and stable operation.

- **Microcontroller:** Raspberry Pi 4
- **Camera Module:** Raspberry Pi Camera Module or compatible USB camera
- **Sensors (Optional):** Motion/PIR sensors for movement detection
- **Indicators (Optional):** LED lights for local detection signals
- **speakers**
- **Power Supply:** 5V DC adapter or battery system or solar panel
- **Programming Language:** Python
- **Libraries/Frameworks:** OpenCV, TensorFlow Lite
- **Communication Platform:** Blynk API for real-time notifications

Project Theme (tick one if related):

- [] SYSTEM (web based application / mobile apps with database)
[] MULTIMEDIA (AR, VR, e-learning mobile apps without database)
[/] HYBRID (IoT)

****Storyboard (Attach as Appendix)**

****Project Milestone/Gantt Chart (Attach as Appendix)**

SECTION C: ACKNOWLEDGEMENT

The stated project proposal has been agreed by the whole group to proceed as Diploma Project I (DAT 21101) and I admit the idea is our team's own idea.

Date : 13 November 2026
Signature : 
Name : Dannie Ezraf Bin Badrol Hsan

SECTION D: SUPERVISOR ENDORSEMENT (FOR SUPERVISOR)

The stated project proposal has been endorsed and agreed to proceed as Final Year Diploma Project I (DAT 21101)

Fauziah Binti Abdul Razak
Pensyarah
Jabatan Teknologi Maklumat
Centre for Diploma Studies (CeDS)
Universiti Tun Hussein Onn Malaysia
[Name], Supervisor Stamp

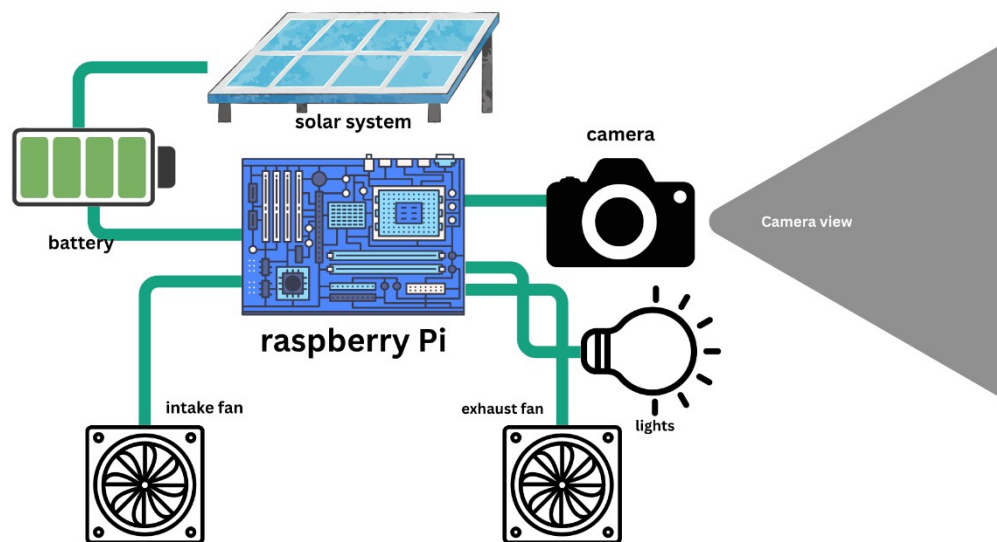
SECTION E: APPROVAL (FOR OFFICE USE)

Comments:

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- ☐ Approved
☐ Rejected
☐ Others [Specify]

Storyboard



animals are detected by the camera



blinking lights will alert driver to be careful
of surrounding area

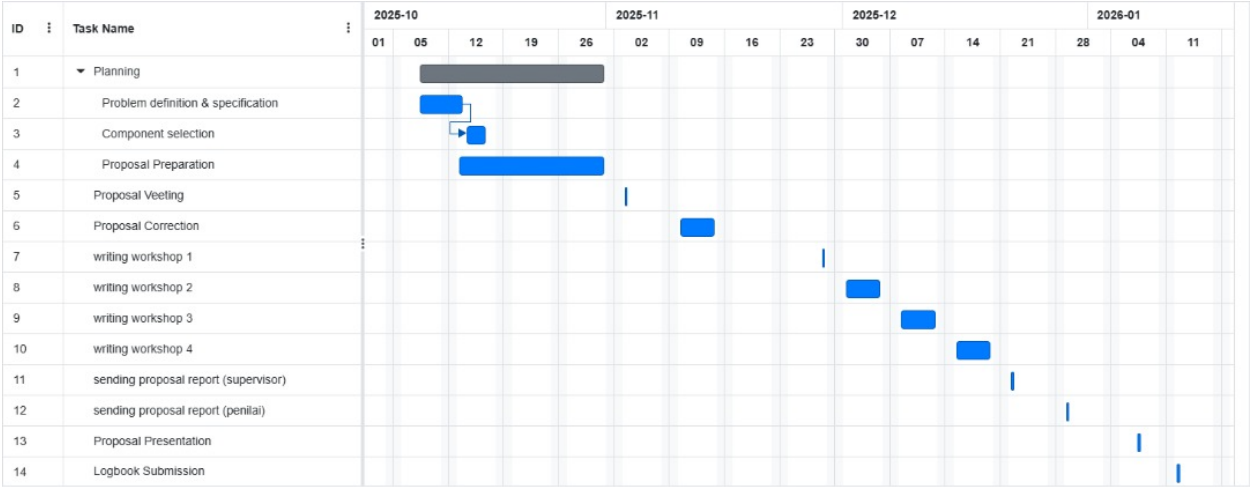


image of the wild animal will be taken
by the camera



picture will be sent to the Police via Blynk

Gantt Chart



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